

From Rubbish to Resource: Waste-Stream-Specific Scenario Narratives for the Future of Secondary Raw Material Recovery in the EU

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Abstract

Securing the future supply of raw materials—particularly critical raw materials (CRMs)—is essential to sustaining economic and societal well-being. Effective forecasting of this supply requires consideration of secondary raw materials (2RMs) contained in diverse waste streams, alongside primary raw materials (1RMs) extracted from the environment. Given the deep incertitude inherent in fortune-telling, it is prudent to explore a broad range of possible developments.

This paper presents the narrative construction phase of a scenario development process that explored future pathways for waste management and material recovery up to 2050. These storylines were developed within the Horizon Europe-funded project FutuRaM, which aims to establish methodology, reporting, and guidance to strengthen the raw materials knowledge base within the EU and beyond.

Three scenarios—Business as Usual (BAU), Recovery (REC), and Circularity (CIR)—were constructed to reflect divergent political, economic, and technological trajectories. Each scenario consists of an overarching narrative, complemented by detailed qualitative projections regarding the future composition and volume of specific waste categories. The scenario development process employed a structured approach integrating forecasting and backcasting techniques, identification of key drivers and uncertainties, extensive stakeholder engagement, and the creation of coherent narrative storylines grounded in current trends, stated policy goals, and published forecasts.

These narratives can provide a valuable foundation for the quantitative modelling of waste generation and material recovery potentials using methods such as life cycle assessment (LCA) and material flow analysis (MFA). In doing so, their utility for decision-making and strategic planning can be further enhanced.

Keywords

Circular Economy, Scenario Development, Storyline Narratives, Backcasting and Forecasting, Waste and Resource Management, Secondary Raw Materials, Critical Raw Materials, Stakeholder Engagement

List of abbreviations

1RM	Primary Raw Material
2RM	Secondary Raw Material
BATT	Waste Batteries (waste stream name)
BAU	Business-as-usual (scenario name)
CDW	Construction and Demolition Waste (waste stream name)
CE	Circular Economy
CIR	Circularity (scenario name)
CRM	Critical Raw Material
EEE	Electrical and Electronic Equipment
ELV	End-of-Life Vehicles (waste stream name)
EoL	End-of-Life
EoU	End-of-Use
EoW	End-of-Waste
EPR	Extended Producer Responsibility
EU	European Union
EU27+4	EU + Iceland, Norway, Switzerland, and the UK
EV	Electric Vehicle
FutuRaM	Future of Raw Materials (EU Horizon 2020 project)
GDP	Gross Domestic Product
IAM	Integrated Assessment Model
ICE	Internal Combustion Engine
IEA	International Energy Agency
IRTC	International Round Table on Materials Criticality
LCA	Life Cycle Assessment
LIB	Lithium-ion Battery
LMT	Light Means of Transport
MFA	Material Flow Analysis
MIN	Mining Waste (waste stream name)
NIMBY	Not In My Backyard (parochial protectionism)
RCP	Representative Concentration Pathway
REC	Recovery (scenario name)
REE	Rare Earth Elements
re-X	A broad set of circular economy strategies (“reduce”, “reuse”, “repair”, “recycle” etc.)
SLASH	Slags and Ashes (waste stream name)
SRM	Strategic Raw Material
SSP	Shared Socioeconomic Pathway
TRL	Technology Readiness Level
UK	The United Kingdom of Great Britain and Northern Ireland
UN	United Nations

UNFC	United Nations Framework Classification for Resources
WEEE	Waste Electrical and Electronic Equipment (waste stream name)
WFD	Waste Framework Directive

1. Introduction

1.1. Context and motivation

Europe's material economy depends on a secure and sustainable supply of raw materials. Primary raw materials (1RMs) extracted from virgin resources underpin manufacturing, energy, mobility, and digital infrastructure, but their extraction and processing entail significant environmental and social impacts and are exposed to geopolitical and market risks. Secondary raw materials (2RMs)—materials recovered from products, components, and residues at end-of-life—offer a complementary source that can reduce dependence on 1RMs, lower life-cycle burdens, and support circular-economy objectives. Within the broader category of raw materials, critical raw materials (CRMs) are those with high economic importance and elevated supply risk; ensuring their availability is central to decarbonisation and digitalisation strategies in the EU.

The availability and recoverability of 2RMs depend on dynamic, interlinked drivers: product design and durability; collection, sorting, and recycling infrastructure; market acceptance and quality of secondary outputs; energy systems; trade and industrial policy; and consumer behaviour. These factors evolve under uncertainty on multi-decadal time horizons and interact across sectors and policy domains. As a result, decision-makers require forward-looking analyses that can systematically explore alternative futures, illuminate trade-offs, and test the implications of policy and technology choices.

Scenario analysis provides such a lens and is widely used in environmental and resource domains to investigate long-term developments under uncertainty (Bishop et al., 2007; Dreborg, 1996; Córdova-Pozo et al., 2023; Sardesai et al., 2021; Amer et al., 2013; Börjeson et al., 2005; Skea et al., 2021; van Notten et al., 2003). While numerous scenario sets exist for energy systems and mobility transitions at national, EU, and global scales, there remains a shortage of scenarios explicitly centred on 2RM/CRM recovery systems and the waste streams that supply them. Addressing this gap, the Horizon Europe project FutuRaM develops a knowledge base on the availability and recoverability of 2RMs within the EU, with a specific focus on CRMs, to enable fact-based decision-making and transparent dissemination of data and methods (FutuRaM Consortium, 2022a; FutuRaM Consortium, 2022b).

FutuRaM focuses on six waste streams that have great potential for 2RM, and CRM supply: batteries (BATT), waste electrical and electronic equipment (WEEE), end-of-life vehicles (ELV), construction and demolition waste (CDW), mining residues (MIN), and slags and ashes (SLASH). The project integrates material stock-flow information, considers economic, technological, regulatory, social, environmental, and geopolitical drivers, and aligns project outputs with the United Nations Framework Classification for Resources (UNFC) (UNECE, 2023). Against this backdrop, the present paper reports the narrative construction phase of FutuRaM's scenario development, which underpins subsequent quantification and modelling.

1.2. Scenario types and their use in this study

In futures studies, scenario exercises are commonly grouped into three families that address different questions (Bishop et al., 2007; Córdova-Pozo et al., 2023; Skea et al., 2021; Amer et al., 2013; Börjeson et al., 2005):

- **Predictive** — *What will happen?* These scenarios explore likely developments given current conditions and stated policies, typically using forecasting assumptions. They are informative for baseline planning but are contingent on assumptions and may under-represent low-probability disruptions.
- **Explorative** — *What could happen?* These map a space of plausible futures by varying external drivers and internal responses. They support preparedness across a range of outcomes without attaching probabilities or preferences.
- **Normative** — *How can a specified target be reached?* These work backwards from desired end-states (e.g., policy targets) to identify feasible pathways and enabling measures. They are purpose-oriented and depend on the choice of goals.

In practice, robust scenario work often combines these families and integrates qualitative narratives with quantitative parameters (Bishop et al., 2007; Amer et al., 2013). This study adopts a mixed approach tailored to 2RM/CRM systems.

The three scenarios presented in this paper are designed to be internally consistent, comparable across the six waste streams, and sufficiently specified to support later parameterisation and modelling.

1.3. Policy context for 2RM/CRM scenarios

EU strategies and legislation provide both direction and constraints for 2RM/CRM systems. The European Green Deal sets the overarching objective of climate neutrality by 2050 and promotes resilient, sustainable industrial value chains (European Commission, 2019). The Circular Economy Action Plan outlines measures to improve resource efficiency, product durability and repairability, waste prevention, and high-quality recycling, with close links to the EU Industrial Strategy (European Commission, 2020). Together, these frameworks create strong signals favouring design for circularity,

increased recovery rates, and market development for 2RMs.

The Critical Raw Materials Act introduces specific supply-security benchmarks: by 2030, not more than 65% of annual EU consumption of each strategic raw material should stem from a single non-EU country; within the EU, capacities should be sufficient to extract $\geq 10\%$, process $\geq 40\%$, and recycle $\geq 15\%$ of annual consumption (European Union, 2023). These benchmarks inform the REC and CIR scenarios, which examine the technology- and system-level changes needed to approach them via domestic recovery and circularity measures.

Waste-stream-specific legislation is also pertinent. BATT, ELV, and WEEE are governed by dedicated directives and regulations that set collection, treatment, and recycling requirements (European Union, 2003; European Union, 2012a; European Union, 2012b; European Union, 2020; European Union, 2023a; European Union, 2023b). These instruments shape the feasible ranges of collection rates, treatment efficiencies, and design obligations used in the scenarios. Notably, several statutory targets—particularly in batteries—are ambitious to the point that meeting them would require rapid structural reforms and performance improvements across the recovery chain. Accordingly, BAU does not assume full attainment of these targets within the study horizon, whereas REC and CIR explore pathways that deliberately mobilise technological and systemic measures to meet or approach them.

A non-exhaustive list of relevant policies that were considered in this study are presented in appendix A3

1.4. FutuRaM aim, scope, and contribution of this paper

FutuRaM's overarching aim is to develop a transparent knowledge base on 2RM availability and recoverability in the EU, with special attention to CRMs, to support evidence-based decisions by industry, investors, and policy-makers (FutuRaM Consortium, 2022a; FutuRaM Consortium, 2022b). The project:

- focuses on the EU-27 plus Iceland, Norway, Switzerland, and the UK (EU27+4) over 2025–2050;
- addresses six waste streams: BATT, WEEE, ELV, CDW, MIN, SLASH;
- integrates data on stocks, flows, technologies, and markets;
- considers interactions with manufacturing, energy, and transport systems as well as environmental, economic, social, technological, regulatory, and geopolitical drivers; and
- aligns outputs, where appropriate, with UNFC classification.

Within this programme, Work Package 2 (WP2) develops a set of plausible scenarios that (i) characterise future 2RM supply from the six waste streams, (ii) identify key drivers and uncertainties, and (iii) provide a structured basis for quantification and model coupling. The scenarios are intended to inform both public and private decisions by enabling systematic assessment of policy and technology options and their implications for 2RM/CRM availability and related environmental-economic outcomes. Selected project needs and actions are summarised in Table X of the manuscript.

This paper reports the scenario narrative development and presents the qualitative storylines. Specifically, it: (1) synthesises policy, technology, and market evidence with expert judgement to build internally consistent storylines for BAU, REC, and CIR across the six waste streams; (2) documents the rationale for the chosen scenario typology and the treatment of EU targets and benchmarks; and (3) provides a traceable basis for subsequent parameterisation and quantitative modelling within FutuRaM. In doing so, it addresses the current gap in 2RM/CRM-centred scenario work by offering a coherent framework and transparent assumptions tailored to waste-derived material supply.

2. Methodology

This section outlines the scenario development workflow applied in FutuRaM (Figure 1). We detail Steps 1–7; parameterisation of scenario variables (Step 7) is reported in Appendix 6, while quantitative modelling and implementation (Steps 8–9) fall outside the scope of this study. Selected outputs from Step 4 are presented in the Results (subsection 3.1) and supporting appendices (A5).

2.1. Step 1: Establish framework and scope

Framework. Scenarios were designed to ensure scientific rigour, transparency, and policy relevance. They are (i) evidence-based and methodologically robust; (ii) coherent and internally consistent while spanning a range of plausible futures and key uncertainties; (iii) decision-oriented, clarifying potential consequences of alternative choices; (iv) participatory, drawing on domain experts and stakeholders; (v) transparent and well documented to enable replication and open dissemination; (vi) adaptive, with provisions for periodic updating; (vii) systems-oriented, capturing interconnections among sectors, waste streams, and policy domains; (viii) long-term in perspective; (ix) integrative of quantitative modelling and qualitative narratives; and (x) communicated clearly to support both practitioner uptake and scholarly contribution. To facilitate consistent communication, a shared vocabulary was established at the outset (Appendix A1).

Scope. The scenarios examine plausible futures for waste management, resource recovery, and the circular economy across the EU-27 plus Iceland, Norway, Switzerland, and the UK (EU27+4) over 2025–2050, with projection years at 2025, 2030, 2035, 2040, 2045, and 2050. Temporal granularity was limited to avoid spurious precision given long-term uncertainty (Amer et al., 2013; Spaniol and Rowland, 2017; Kahn and Wiener, 1967). The analysis centres on six FutuRaM waste streams—WEEE, ELV, BAT, CDW, MIN, and SLASH—together with interfaces to manufacturing, energy, and transport systems, and the relevant environmental, economic, social, technological, and geopolitical policy domains.

2.2. Step 2: Determine methodology

Selection criteria and methods considered. We identified methods and tools suited to FutuRaM’s objectives using criteria of relevance and applicability to context; feasibility (time, budget, expertise, data); transparency and documentation; flexibility to incorporate new evidence; accessibility for non-experts; effectiveness and efficiency; and stakeholder acceptability. Candidate methods are summarised in Appendix A2.

Core scenario set and candidate dimensions. The grant proposal defined three core scenarios—Business as Usual (BAU), Recovery (REC), and Circularity (CIR)—to explore how secondary raw material (2RM) availability is shaped by the evolution of 2RM recovery systems and re-X strategies. During scoping, additional dimensions were considered, notably supply-chain security and the pace of the energy transition. Their inclusion was tested using the procedures below. The reasons for ultimately not adopting further dimensions are discussed in Section 4.

Multi-criteria and cross-impact analysis. To avoid uncontrolled growth in scenario variants, we applied a matrix-based multi-criteria and cross-impact analysis to evaluate internal consistency and plausibility of combinations (Steubing et al., 2016). For instance, “low” progress in the energy transition was judged inconsistent with “high” progress in recycling/circularity indicators and could be excluded. By contrast, supply-chain security varied more independently, but credible geopolitical speculation was not feasible; incorporating it as a scenario dimension would introduce excessive uncertainty. Instead, potential supply constraints will be analysed via sensitivity tests and, where appropriate, explorative multi-objective optimisation to address questions such as: *What happens to the 2RM system if element x is constrained, and what responses are optimal?* Additional rationale is provided in the Discussion.

Delphi process. Early stages of scenario building were supported by a Delphi process to aggregate expert judgement across waste streams and recovery-system domains (Hsu et al., 2019). Steps included expert selection, initial questionnaires, iterative rounds with controlled feedback, and convergence toward consensus. Subsequent rounds will include external stakeholders from industry, academia, and government.

2.3. Step 3: Marker-scenario mapping

We conducted a structured literature review to identify and analyse marker scenarios relevant to FutuRaM. This mapping served to (i) build on established knowledge; (ii) benchmark assumptions, drivers, and methodological approaches; (iii) identify gaps; and (iv) enhance comparability and credibility of our scenarios. Insights from marker scenarios informed the selection of drivers, the structuring of storylines, and the alignment of projection years and sectoral linkages. The consulted marker scenarios are listed in Appendix A4.

2.4. Step 4: Identify key drivers and select scenario elements

Purpose and approach. This step identified the social, economic, technological, environmental, and policy factors that shape the waste-management system over time. The process combined literature review, internal expert elicitation, and stakeholder engagement to assess trends, uncertainties, and emerging issues. It was iterative, with feedback loops to ensure

coverage of the most influential drivers and to maintain consistency with scope and objectives.

Sub-method and workflow. Figure 2 depicts the sub-method for identifying and defining scenario elements:

- **Preliminary collection and screening.** Candidate elements were compiled from peer-reviewed literature, statistics, policy documents, and project datasets, then screened against predefined criteria for relevance, reliability, plausibility, and model utility. Overlapping items were consolidated (e.g., recyclability mandates with design-for-recyclability), and elements judged irrelevant, non-modellable, or unreliable (e.g., corruption, data protection, supply-chain conflict) were excluded.
- **Assessment and categorisation.** Retained items were organised by thematic relevance and characteristics, yielding 21 key elements (Table 1). While CIR and REC share many elements, they differ in how targets are achieved: CIR emphasises re-X strategies (reduce, reuse, repair, remanufacture, recycle), whereas REC prioritises advances in recovery technology and infrastructure—an important distinction for subsequent quantitative modelling.
- **Scope classification.** Each element was classified according to its relationship to the waste-management system: **internal** (within scope for detailed research and modelling, e.g., composition and design changes), **external** (included in storylines, varying over time but constant across scenarios, e.g., population, GDP), or **outside** (excluded from scenarios but, where relevant, examined in sensitivity analysis, e.g., supply constraints).

Selected Step-4 outputs are presented in the Results and relevant appendices. The validated element set informed the design of storyline themes (Step 5) and the drafting of narratives (Step 6).

2.5. Step 5: Develop storyline themes

The overarching scenario themes were guided by the FutuRaM charter and the grant proposal's core scenario set. Themes focus on how 2RM availability evolves under different combinations of recovery-system development and re-X strategies. The three scenarios are:

- **Business as Usual (BAU)** — *Predictive*: projects outcomes if current policies and trajectories persist, assuming continuity in waste management and resource recovery systems.
- **Recovery (REC)** — *Normative*: prioritises technological and infrastructural improvements in recovery systems to meet EU material-use and recovery targets.
- **Circularity (CIR)** — *Normative and explorative*: aligns with the EU Circular Economy Action Plan, examining systemic adoption of re-X strategies to achieve targets and reduce dependence on primary resources.

A graphical depiction of the scenarios is provided in Figure 3. The rationale for not expanding the scenario set with additional dimensions is discussed in Section 4.

2.6. Step 6: Develop qualitative narrative storylines

Narratives were constructed through an iterative process that combined systematic evidence gathering with structured consultation. Data on trends, targets, policies, and marker scenarios were collated to define candidate drivers and elements. These inputs were scrutinised in workshops and Delphi-style rounds with consortium experts and, where appropriate, external stakeholders, to test plausibility, resolve inconsistencies, and prioritise influential factors. Draft narratives for BAU, REC, and CIR were reviewed for internal coherence and alignment with the agreed scope, then refined through successive feedback cycles.

Each storyline integrates qualitative insights with model-ready descriptors to enable later parameterisation. Assumptions, causal links, and uncertainty ranges were documented to support transparency and to flag items for sensitivity analysis. The outputs of Step 6 thus provide: (i) coherent narratives that explain how and why system trajectories diverge across scenarios; (ii) a traceable mapping from qualitative drivers to quantitative variables; and (iii) an auditable record of literature sources, expert judgements, and decisions.

The main outcome of this study, abridged scenario storyline narratives are presented in the Results (subsection 3.2). More details about the waste-stream-specific impacts of the scenarios is presented in Appendix A7.

Forward link to Step 7. Parameter values and time paths for model variables (e.g., technology performance, collection and sorting rates, design and composition shifts) are to be quantified in Step 7 using trends and forecasts, subject to the consistency constraints defined in Steps 2–6. These parameters were analysed in preparation for quantification and reported in Appendix A6. **Steps 8–9** (quantitative modelling and implementation) are outside the scope of this study.

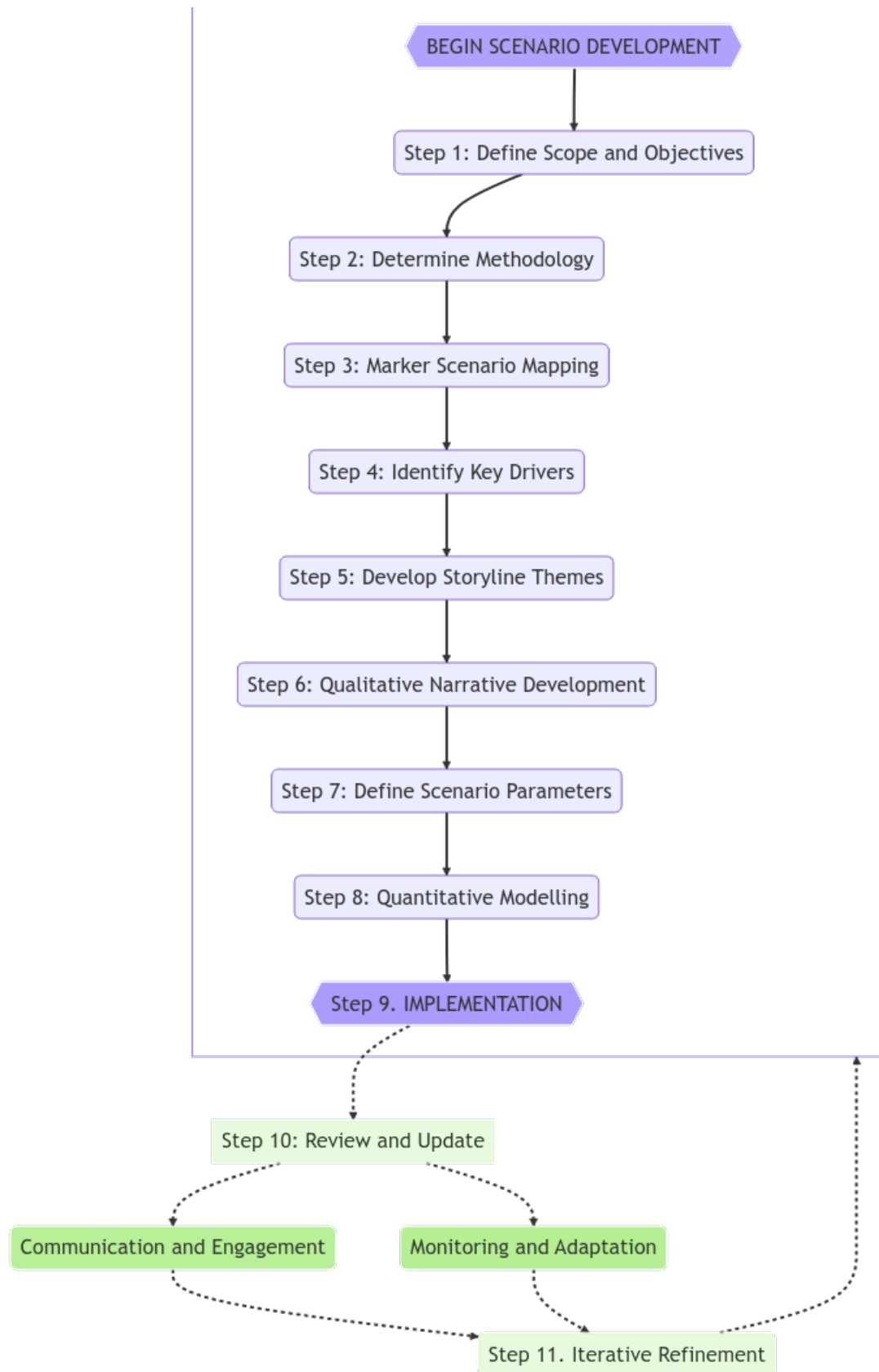


Figure 1. A flowchart representation of the methodology used for the scenario storyline development process in FutuRaM

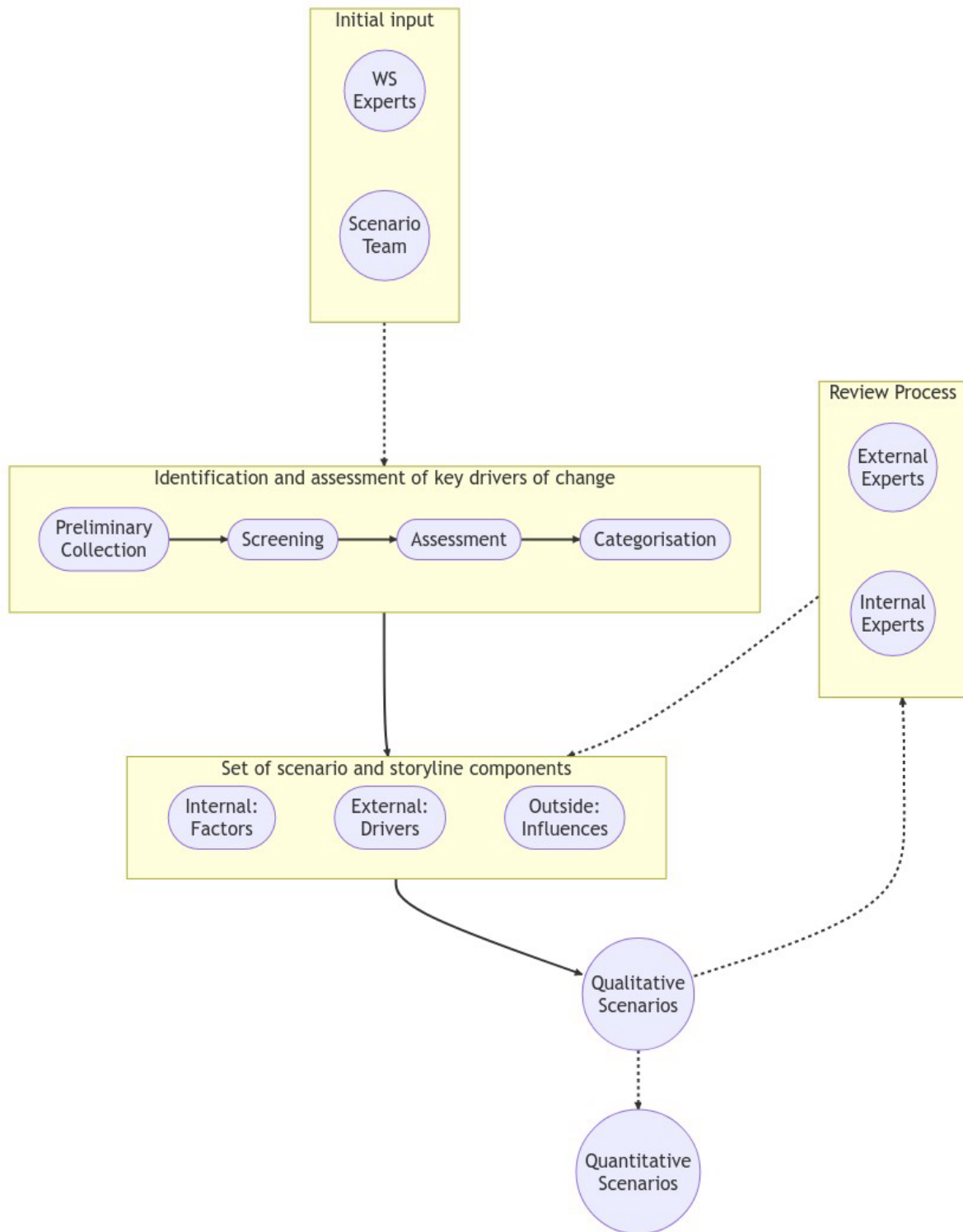


Figure 2. An illustration of the methodology used for identifying, defining, screening, and integrating the key elements of the scenarios developed in FutuRaM

3. Results

3.1. Scenario element selection

Results from the preliminary collection of scenario elements are summarised in Appendix A5. Table 1 lists the drivers and factors retained after the screening process. Appendix A6 presents the final, categorised set of scenario elements and indicates which are carried forward for parameterisation and subsequent modelling. Elements were excluded when judged out of scope, insufficiently specific to the focal waste streams, or infeasible to quantify.

Table 1. Scenario drivers and factors retained after screening. Items were omitted if they were out of scope, not sufficiently specific to the waste streams, or infeasible to quantify. "Scenario variances" denote a qualitative stakeholder assessment of the expected trend for each driver/factor in the three scenarios—Business as Usual (BAU), Recovery (REC), and Circularity (CIR).

Domain	Factor/Driver	Definition	Scenario variance		
			BAU	REC	CIR
Economic	CO ₂ market price	Price of CO ₂ emissions in carbon markets	I	I	I
Economic	Economic growth	Rate of economic growth	I	I	I
Economic	Energy prices	Prices of energy resources	I	I	I
Economic	Market saturation	Level of saturation reached in the market for certain products or services	I	I	II
Economic	Raw material vs 2RM prices	Price comparison between raw materials and Secondary Raw Materials (2RMs)	I	I	I
Economic	Re-industrialisation of EU	Process of revitalizing industrial activities in the EU	I	I	I
Environmental	Climate change impacts	Impacts of climate change such as flooding and other related events	I	I	I
Environmental	Climate change mitigation efforts	Efforts made to mitigate the effects of climate change	I	I	I
Environmental	Increased drive for environmental protection	Growing motivation and drive to protect the environment	I	III	III
Environmental	Resource shortage	Shortage of natural resources	I	I	I
Legal/Political	Ecodesign/re-X mandates	Establishment of ecodesign requirements for specific product groups to improve circularity, energy performance, and other environmental sustainability aspects	I	II	III
Legal/Political	Governance: corruption vs compliance	Contrasting levels of corruption and compliance within governance systems	I	I	I
Legal/Political	International trade and co-operation (vs. autarky)	Level of international trade and cooperation versus self-sufficiency	I	I	I
Legal/Political	Product information transparency	Provision of transparent product information to consumers, manufacturers, importers, repairers, recyclers, or national authorities	I	III	III
Legal/Political	Progress toward renewable energy targets	Progress made in achieving renewable energy targets	I	I	I
Legal/Political	Stricter environmental regulations	Implementation of more stringent rules and regulations for environmental protection	I	III	III
Legal/Political	Subsidies/taxation to promote circularity	Provision of subsidies or implementation of taxation policies to incentivize circularity	I	I	I
Legal/Political	Supply chain due diligence laws	Implementation and enforcement of laws requiring companies to address negative human rights and environmental impacts in their value chains	I	II	III
Social	Hoarding	The act of stockpiling and keeping excessive amounts of products	III	II	II
Social	NIMBY attitudes toward projects	Opposition of local communities to the location of new projects, such as mining, in their vicinity	I	I	I
Social	Participation in re-X activ-	Involvement in activities related to the "re-" concepts,	I	II	III

Domain	Factor/Driver	Definition	Scenario variance		
			BAU	REC	CIR
	ities	including refusing, reducing, repairing, and reusing products			
Social	Population	Size and growth of the population	I	I	I
Social	Urbanisation	Rate of urban population growth	I	I	I
Technical	Digitisation	Adoption and integration of digital technologies	I	I	I
Technical	2RM system integration	Integration of a 2RM system across the EU	I	III	III
Technical	Product technology	Changes in product function or composition that lead to changes in waste stream composition and quantity	I	III	III
Technical	Recovery technology	Technologies and processes for recovering materials from waste	I	III	III

3.2. Qualitative narrative development: The scenario storylines

This subsection presents abbreviated qualitative storylines for the three scenarios—**Business as Usual (BAU)**, **Recovery (REC)**, and **Circularity (CIR)**—summarising their system-wide logic and expected trajectories. Figure 3 illustrates the distinct economic logics embodied by each scenario: trend continuation in BAU, technology-led improvements in recovery systems in REC, and systemic adoption of re-X strategies in CIR.

In this subsection, the abridged scenario storyline narratives are presented. More detailed information about the waste-stream-specific impacts of each scenario is given in appendix A7.

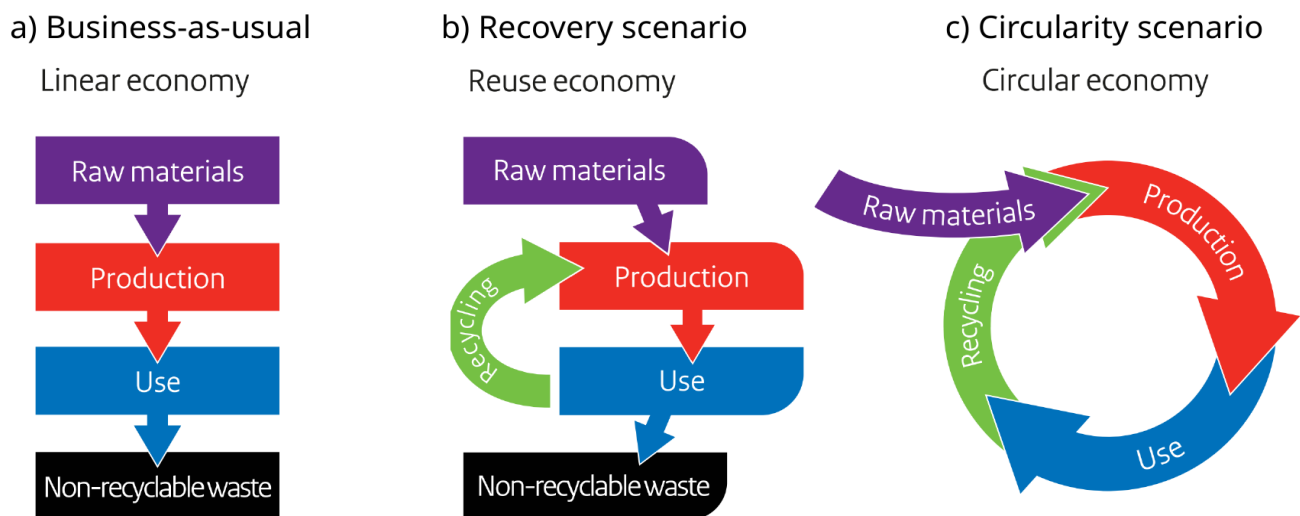


Figure 3. A schematic representation of the three discrete economic models embodied by the three scenarios developed in FutuRaM: a) Business-as-usual (BAU), b) Recovery (REC), c) Circularity (CIR). Adapted from Government of the Netherlands (2016).

3.2.1 Scenario I: Business-as-usual (BAU)

The BAU scenario projects present trends through to 2050 with minimal deviation in consumption patterns and in the organisation of the secondary raw material (2RM) system (iea2022worldenergyoutlook). While incremental gains occur—e.g., moderate resource-efficiency improvements, gradual advances in recovery technology, and a steady energy transition—transformational change is constrained by economic pressures, social preferences, political inertia, and entrenched interests. Primary extraction remains the dominant source meeting the EU's growing material demand.

System trajectory and assumptions

BAU is a forecasting-led pathway: it extrapolates current drivers and institutions, assuming today's policy ambition and enforcement largely persist. Minor efficiency gains are absorbed by rising demand, leaving overall material throughput closely coupled to GDP growth. The linear take–make–dispose model continues to shape production and consumption, and the 2RM system improves only at the margin.

Material demand and supply

Material demand remains tightly linked to economic activity, so total consumption increases over time. Primary mining and extraction continue as the leading supply sources, reinforcing reliance on traditional extraction methods. Base metals benefit from established markets and economies of scale and are comparatively well recycled; by contrast, many rare and specialty metals are still lost because technologies and economics limit their recovery. EU mining remains limited and geographically concentrated in a few member states; known domestic potentials—e.g., lithium (PT, FR, UK) and rare earths (SE)—are not realised. Import dependency therefore escalates, increasing exposure to supply disruptions and market volatility (iea2023crm; mancini2013supplysecurity).

Waste management and recovery performance

Recycling and recovery rates remain low relative to potential, with persistent losses of critical raw materials. Producer responsibility schemes and data availability improve only incrementally; collection performance and contamination continue to limit output quality and value. Investments in sorting, pre-treatment, and recovery capacity are selective and insufficient to change system-level outcomes. As a result, significant 2RM opportunities are missed, and landfilling or low-value outlets remain prevalent in several streams.

Policy and governance

Many EU targets for ecodesign, recycling, recovery, and waste prevention are not met. Regulatory tightening is constrained and uneven across member states, providing weak incentives for high-quality recyclates or design changes. Where policies advance, implementation and monitoring gaps blunt impact. Without stronger and more consistent measures, innovation remains patchy and the cost of secondary outputs often fails to match primary alternatives at required specifications.

Environmental and socio-economic outcomes

Environmental pressures from extraction persist, contributing to ecosystem degradation, biodiversity loss, and climate impacts (ipcc2024climate). Relative to pathways with stronger circular measures, BAU exhibits higher cumulative energy and material footprints. At the same time, heightened import reliance exposes EU industry to price spikes and supply shocks, increasing planning uncertainty and potentially delaying investment in new recovery technologies (mancini2013supplysecurity; iea2023crm). Industrial priorities emphasise short-term cost efficiency over long-term resilience and sustainability. Transitions to renewable energy and e-mobility proceed, but at the present pace rather than on an accelerated trajectory.

Dependencies and risks

Key vulnerabilities include: (i) persistent coupling of demand and GDP; (ii) limited progress on design-for-circularity and product information, which constrains high-quality recovery; (iii) under-investment in 2RM technologies and infrastructure; and (iv) import-related exposure to geopolitical and market risks. These factors reinforce lock-in to linear practices, making it harder to realise future gains without stronger policy and market signals.

Positioning relative to other scenarios

BAU serves as a **reference** against which the Recovery (REC) and Circularity (CIR) scenarios can be evaluated. Compared with REC, BAU exhibits lower recovery performance, weaker output quality, and higher landfilling; compared with CIR, it lacks the upstream design, use-phase, and sufficiency levers needed to decouple services from primary extraction. In short, BAU represents the lower-bound pathway where incremental improvements occur but the structure of the system—and its associated risks—largely persists.

3.2.2 Scenario II: Recovery (REC)

The recovery scenario centres on harnessing pre-consumer and post-consumer waste as a significant source of secondary raw materials (2RMs). It moves decisively beyond the status quo by upgrading recovery performance and infrastructure, yet overall material throughput continues to track growing demand rather than absolute reductions. The modelling approach combines backcasting for dimensions governed by policy targets with forecasting for areas without binding targets, linking desired outcomes to present conditions and projecting feasible pathways from current baselines.

System focus and technological change

Waste treatment is the pivotal actor. The emphasis is on rolling out and optimising advanced recovery technologies to improve sorting fidelity, material purity, and process efficiency. Digitalisation—spanning artificial intelligence, automation, robotics, and enhanced process control—raises plant productivity and stabilises quality, allowing higher-value recyclates to re-enter manufacturing. Investments concentrate on upgrading sorting, pre-treatment, and recovery facilities, and on data systems that disclose product composition and design attributes, which in turn enable better characterisation and targeted recovery. Producer responsibility schemes mature, driving higher take-back and collection across key streams and reinforcing steady feedstock to recyclers.

Policy, governance, and market organisation

EU targets for recycling and recovery are reached due to a more expansive, efficient, and effective waste management system. Regulatory instruments are tightened to promote high-quality outputs (e.g., contamination limits, de-risked offtake), improving the economics of recovery. Standardisation and clearer guidance support EU-wide implementation and compliance; cross-border coordination and knowledge transfer help diffuse good practice. Alongside regulation, public investment and R&D programmes crowd in private capital for plant upgrades and process innovation, while transparency requirements expand the availability of product and material data relevant to recovery planning.

Material supply and risk profile

Higher domestic recovery partially mitigates supply risk by reducing exposure to volatile global markets; nonetheless, import dependence persists for several critical inputs where domestic end-of-life stocks, technology readiness, or economics remain constraining (iea2023crm). Urban mining and recovery from selected waste streams and tailings are explored more systematically, with pilots scaling where yields and costs are favourable. Industrial actors increase the share of secondary inputs where price–quality conditions permit, but adoption varies across sectors and member states.

Environmental and socio-economic outcomes

Relative to business-as-usual (BAU), environmental burdens decline due to higher recovery rates and lower landfilling. However, upstream pressures linked to aggregate consumption growth remain significant—this is a system that recovers more, not one that uses less. Employment expands in recycling, recovery, and associated services as collection improves and plants scale. Cost efficiency in 2RM production improves through learning and quality gains, further reinforcing uptake in manufacturing where specifications can be met.

Diffusion dynamics and behavioural dimensions

Innovation diffusion is uneven. Frontrunner regions and sectors demonstrate higher yields and faster quality improvements; laggards remain closer to BAU performance. Public awareness of recycling improves, but behaviour change is still concentrated at the end-of-pipe (separating and returning products) rather than sufficiency (avoiding or reducing consumption). As a result, the system attains higher circular recovery without fully addressing demand-side drivers.

Dependencies and implementation risks

Realising the scenario's gains depends on sustained investment in recovery infrastructure and digital capabilities; robust enforcement of quality standards; and continued maturation of producer responsibility schemes. Key risks include: (i) inconsistent implementation and monitoring across member states; (ii) quality shortfalls that depress secondary material prices or restrict their use in high-specification applications; and (iii) slower-than-expected technology learning, which would limit cost and performance improvements. These risks are partially offset by clearer regulatory signals, improved data availability on product composition and design, and public procurement practices that accept certified secondary content.

Positioning relative to other scenarios

Compared with BAU, REC delivers higher recovery, lower landfilling, and better-quality secondary outputs through technology, standards, and organisation. In contrast with a full circularity pathway, REC places most of its leverage at end-of-life rather than at design and use; it moderates—but does not decouple—material demand. Consequently, import reliance is reduced only partially, and remains material for certain critical inputs (iea2023crm).

3.2.3 Scenario III: Circularity (CIR)

System trajectory and assumptions

CIR deploys waste prevention, ecodesign, reuse, repair, remanufacturing, and high-quality recycling to the maximum feasible extent, with the explicit aim of decoupling material services from primary resource extraction. The modelling approach combines **backcasting** for factors with binding targets and forecasting for factors without them, linking desired outcomes to present conditions and defining consistent pathways from now to 2050.

Material demand and supply (towards decoupling)

Relative to BAU and REC, total material throughput is moderated by upstream measures (design for longevity, sharing and service models) and higher 2RM use. 2RM recycling and recovery rates rise markedly; secondary materials gain wider acceptance in production as standards and data improve. Reliance on imports declines as domestic circular flows expand, although strategic procurement remains important in specific value chains. Renewable energy and cleaner process technologies are applied throughout 2RM production and use to support a low-carbon trajectory.

Waste management and recovery performance

End-of-life systems shift from volume handling to quality-focused recovery. Standardised collection, advanced sorting, and traceable feedstocks (e.g., product passports, waste tracking) raise output purity and value, enabling high-quality, loop-

closing recycling. Reuse, repair, refurbishment, and remanufacture absorb a larger share of post-consumer inflows; residual fractions are more readily segregated and recovered at high efficiency. Investments prioritise design-compatible recovery, contamination control, and certification that unlock secondary inputs for high-specification applications.

Policy and governance

EU targets for recycling, recovery, ecodesign, and waste prevention are met, supported by stricter product-design guidelines, robust implementation, and monitoring. Regulatory instruments reward durability, reparability, standardisation, and disassembly; producer responsibility schemes are extended to drive high collection and take-back. Digital information requirements (e.g., product passports) become baseline practice across value chains, improving planning and verification. New business models—product-service systems, leasing, and take-back—scale and moderate traditional consumption patterns (geissdorfer2020circbusinessmodels). Fiscal reform complements regulation (e.g., shifting taxation from labour to resource depletion; flexible VAT rates for repair), while phasing out environmentally harmful subsidies. Member States implement EU measures and tailor complementary regional initiatives; international coordination supports alignment with broader commitments (un2015sdg).

Mechanisms of resource efficiency

CIR enhances resource efficiency via four complementary strategies:

1. **Narrowing** material flows through reduced or shared product use and improved manufacturing efficiency;
2. **Slowing** flows by extending product and component lifetimes via reuse, repair, refurbishment, and remanufacture;
3. **Closing** loops through high-quality recycling that reduces landfilling and incineration and lowers demand for primary inputs;
4. **Substituting** finite inputs with renewable or lower-impact alternatives where feasible.

Design, behaviour and information

Re-X strategies are embedded from the design phase onward (repairability, remanufacturability, standardised components, easy disassembly). Predictive maintenance and monitoring extend service life in sectors such as transport and manufacturing. Transparency tools—digital product passports and waste tracking—standardise information on composition, durability, and reparability, enabling efficient disassembly, reuse, and high-quality recycling, and fostering collaboration along value chains. Public awareness and education increase participation in repair, reuse, and service models, reinforcing demand for long-lived, upgradable products.

Environmental and socio-economic outcomes

CIR retains product, material, and resource value in the economy for longer while minimising waste generation, consistent with a sustainable, low-carbon, resource-efficient, and competitive EU economy. Anticipated effects include:

- **Economic:** greater resilience to resource scarcity and price volatility; emergence of efficient production and consumption models; new business opportunities; diversified employment in circular activities.
- **Environmental:** lower cumulative energy use and mitigation of irreversible ecological damages across climate, biodiversity, and pollution pressures.
- **Social:** broader participation in repair, reuse, and service models; enhanced inclusion via local circular enterprises and skills development.

Dependencies and enablers

Delivery depends on: (i) sustained investment in design-compatible recovery infrastructure and digital information systems; (ii) consistent enforcement of product-design and recovery standards; (iii) mature producer responsibility with verified performance; and (iv) stable market pull for certified secondary content (including public procurement). These enablers reduce quality risk, improve economics for 2RMs, and support continued diffusion of circular practices.

Positioning relative to other scenarios

Compared with BAU, CIR achieves substantially higher recovery quality and volume and moderates demand via upstream design and behavioural levers. Compared with REC, CIR shifts the centre of gravity upstream—placing design, durability, and service models alongside end-of-life performance—thereby advancing from “recover more” to “use better and less,” with stronger progress towards decoupling.

4. Discussion

What this study adds

This paper addresses a clear gap: while many scenario sets exist for energy and mobility, few centre on the waste-derived supply of raw materials and its implications for critical raw materials (CRMs). We contribute a transparent, EU27+4–focused scenario framework with three internally consistent storylines—Business as Usual (BAU), Recovery (REC), and Circularity (CIR)—that are explicitly grounded in waste-stream realities. The narratives integrate qualitative evidence (policies, technology trajectories, market drivers) with model-ready descriptors, creating a traceable bridge from storylines to subsequent parameterisation and quantitative modelling.

Interpreting the three storylines

BAU offers a predictive baseline of incremental change, useful as a reference for policy appraisal. REC is a normative pathway that concentrates on technology and infrastructure in collection, sorting, and processing to raise recovery performance and approach statutory targets for 2RMs relevant to CRMs. CIR is a normative/explorative pathway that layers demand-side and design-side levers—re-X strategies such as reduce, reuse, repair, remanufacture, recycle—on top of improved recovery, aiming to decouple material services from primary raw material (1RM) extraction. Taken together, the trio delineates a wide but decision-relevant envelope of futures for 2RM supply, enabling systematic testing of how policy and technology choices propagate through the resource-recovery system.

Policy relevance: focusing on controllable levers

A key design choice was to prioritise variables that decision-makers can influence (“policy knobs”), including: product design requirements (durability, reparability, standardisation), collection and sorting performance, recycling yields and quality, end-market development for 2RMs, and enabling governance (standards, reporting, extended producer responsibility). By structuring the scenarios around these levers, the narratives remain directly interpretable for regulators, industry, and investors and are well suited to downstream tools such as life cycle assessment and material flow analysis.

Why some dimensions were excluded—and how we still address them

We considered adding explicit axes for supply-chain security and the pace of the energy transition. Both are highly salient for CRMs but proved problematic as core scenario dimensions. Credible trajectories for supply-chain security would require geopolitical forecasting beyond the evidence base and would inflate uncertainty and scenario count without commensurate insight. Similarly, varying the energy transition orthogonally to BAU/REC/CIR created internal inconsistencies (e.g., “low transition–high circularity” combinations) and a combinatorial expansion that diluted the focus on waste-system levers. Instead, we treat these influences through targeted sensitivity analysis and optional module coupling. For example, exogenous supply constraints can be imposed in shock tests, and simplified price modules can probe how alternative 1RM–2RM price paths shape collection and processing responses. This approach preserves clarity and decision usefulness while acknowledging real-world volatility.

Strengths of the approach

Methodologically, three features strengthen credibility and transferability. First, we used marker-scenario mapping to align assumptions and highlight gaps relative to established scenario families, improving comparability. Second, cross-impact and multi-criteria checks were applied to test internal consistency and to reduce false-plausible combinations before narrative drafting. Third, structured expert elicitation (Delphi-style) and stakeholder workshops were used to challenge assumptions, surface disagreements, and converge on the most influential drivers. The result is a set of storylines that are both transparent—assumptions and rationales are documented—and practicable—parameters and projection years are specified to enable quantification.

Limitations and caveats

Several limitations merit attention. The geographical focus (EU27+4) supports policy relevance but constrains generalisability to other institutional contexts; transfer will require careful adaptation of drivers and policy instruments. The time horizon (2025–2050) and discrete projection years balance tractability and detail but cannot capture short-term shocks or continuous dynamics; we caution against over-interpreting mid-point values. Data gaps persist for some waste streams and sub-components (e.g., informal flows, secondary output quality), and expert judgement was necessary in places; while triangulated, expert elicitation carries bias risk. Finally, the present contribution covers narrative construction; full insight into performance trade-offs depends on subsequent parameterisation (Step 7) and quantitative modelling (Steps 8–9), which are outside this paper’s scope.

Implications for modelling and evidence needs

The narratives provide clear cues for model coupling. REC and CIR, in particular, specify parameter families (collection rates, sorting efficiencies, process yields, design shifts) and boundary conditions (policy targets, technology readiness) that can be translated into time paths and tested against constraints. Two priorities emerge: (i) systematic sensitivity analyses

for supply constraints, price dynamics, and technology learning rates; and (ii) selective coupling to market and trade modules to capture feedbacks that materially affect 2RM uptake. Progress also depends on improving evidence for product composition, durability, and design trends, and on better quality metrics for secondary outputs—areas where targeted data collection would materially reduce uncertainty.

Outlook and use

We view the narratives as living artefacts. They can be updated as legislation, technology, and markets evolve, while preserving a consistent backbone for tracking change. Practically, the storylines support three near-term uses: (1) screening of policy packages for internal coherence (e.g., aligning design mandates with recovery infrastructure timelines); (2) investment planning for collection, sorting, and high-quality recycling capacity; and (3) prioritisation of R&D on recovery technologies and circular-design practices most consequential for CRM-relevant 2RMs. By keeping the scenarios centred on controllable levers and by documenting assumptions transparently, the framework aims to enable robust, evidence-informed decisions amid persistent uncertainty.

5. Conclusions

This paper has developed three internally consistent narrative scenarios—Business as Usual (BAU), Recovery (REC), and Circularity (CIR)—to explore how waste management and recovery systems could shape the availability of secondary raw materials (2RMs) relevant to critical raw materials (CRMs) in the EU27+4 to 2050. The storylines integrate policy, technology, and market evidence with structured expert input, marker-scenario mapping, and cross-impact checks, yielding transparent, model-ready descriptors and projection years suitable for subsequent quantification.

A deliberate design choice was to centre the scenarios on levers that decision-makers can influence—product design and standards, collection and sorting performance, recycling yields and quality, and end-market development for 2RMs—rather than on highly exogenous axes. Proposed dimensions such as supply-chain security and the pace of the energy transition were therefore not adopted as core axes; instead, they will be addressed through sensitivity analysis and optional coupling to market/trade and optimisation modules. This preserves interpretability while acknowledging genuine sources of volatility.

The narratives now provide a concrete basis for parameterisation and for the development and testing of quantitative models. Priority tasks include (i) translating storyline cues into time paths for collection, sorting, process efficiency, and design shifts; (ii) systematic sensitivity tests for supply constraints and 1RM–2RM price dynamics; and (iii) targeted data improvements on product composition, durability, and secondary-output quality.

While results are scoped to the EU27+4 and a 2025–2050 horizon, the framework is transferable with appropriate localisation. By focusing on controllable levers and documenting assumptions clearly, the scenarios are intended to support evidence-informed policy and investment decisions under deep uncertainty.